Can Youtube Live Chat Comments be a proxy for fan votes in IDL

Introduction

In recent years, social media platforms such as YouTube, Instagram and TikTok have significantly increased the viewership, exposure and global reach of dance. The International Dance League (IDL), the first professional global dance league, drew billions of views in its debut 2026 season and signed broadcast deals with ESPN2 (International Dance League, 2026c).

The league consists of 6 pro dance teams competing across 5 regular season events and a Championship.

Each event has two rounds: in Round 1, the 6 teams compete in three head-to-head matches; in Round 2, the three Round 1 winners compete in a three-way match (International Dance League, 2026a).

A team's score combines scores from a six-judge panel on ten categories with a fan vote component. In Round 1 matches, each judge awards 1 point to the team they scored highest and 1 fan vote point is awarded to the fan vote winner. The team with most points wins the match. In Round 2, each team's judge score is the average of the 6 judges' total scores and a share of 1 point, proportional to their fan vote share, is added to their judge score to produce a final score. Teams are ranked 1st, 2nd, and 3rd by this final score. The Championship has seven judges and no fan vote component.

Season standings are based on cumulative points earned across all events. At each event, a team earns points from its Round 1 match outcome and its Round 2 ranking, with the top three teams in Round 2 receiving 7, 5, and 3 points respectively. These standings determine seeding for the Championship and count toward team payouts (International Dance League, 2026a).

Fan votes are collected through the IDL web application, with voting open to both in-venue attendees and YouTube livestream viewers. Voting opens after all teams in a match have performed and remains open for a 60 second window (International Dance League, 2026a).

The IDL's format allows fan votes and judges' scores to be compared directly. Fans vote immediately after watching the same routines the judges score, and because every team competes at every event, these comparisons can be tracked for each team across the full season. Additionally, YouTube livestream viewers leave time-stamped comments, providing a real-time record of fan reactions to each routine.

This paper aims to measure the agreement between fan votes and judges' scores, identify patterns in fan voting, and assess whether YouTube livestream chat sentiment reflects fan vote share and judges' scores. It also discusses implications for teams, in how they can connect with both fans and judges, and for the league, in how it designs its scoring.

Relavant Prior Work

Fan votes are used in major sports leagues to select players for all-star games. Kunz-Kaltenhäuser et al. (2023) compared voting patterns for the NFL's All-Pro awards (expert voting) and Pro Bowl selections before and after public balloting was introduced. The study used panel probit regression to model each selection as a function of performance markers (e.g., touchdowns, yards, sacks) and non-performance markers (e.g., age, previous awards, team win-loss record). In MMA, fans can score a bout after it ends using the same 10-point scorecard as judges. Berthet (2024) used percent agreement to measure how closely judges' standard scores and consensus scores aligned with these fan scorecards. Fan votes have also been used to directly influence competition outcomes. Formula E's FanBoost, used from 2014 to 2022, allowed fans to vote for their favourite drivers before and during the early minutes of each race (e-formula.news, n.d.). The drivers with the most votes received a short burst of extra power to use in the second half of the race. Formula E dropped FanBoost for the 2023 season after teams and promoters unanimously agreed to remove it, as part of a broader effort to make the competition more authentic (The Race, 2022). The CEO of the league acknowledged that fans tended to vote for well-known drivers and championship contenders (Autosport, 2018). To the best of our knowledge, sports literature has not yet examined judge-fan agreement, or whether fan votes reflect popularity or performance, in a live competition where fan votes directly affect match outcomes and season standings.

Sentiment analysis is widely used to study sport fans. Chang (2019) used a lexicon-based sentiment classifier to classify real-time tweets posted during the Super Bowl 50 game into discrete emotion categories which aligned with live game scoring events. Positive emotion word counts increased after a fan's team scored and negative emotions word counts increased after the opposing team scored. Kobs et al. (2020) built lexicon-based sentiment classifiers for Twitch-specific slang and emotes and used their outputs to weakly train a neural network, which they applied to plot sentiment over time for a live-streamed game convention keynote, showing a sharp sentiment drop when the audience learned that the newly announced game, Diablo Immortal, would be released for smartphones only. However, these studies interpret sentiment by linking it to events during the game or stream, such as a team scoring; whether sentiment agrees with an independent fan vote outcome has not been studied in a sports context.

No formal research on the International Dance League currently exists. IDL's format allows judges' scores, an official fan vote, and YouTube live chat sentiment to be observed simultaneously for the same performance. Therefore, our paper addresses this gap by measuring agreement between fan votes, judges' scores, and live chat sentiment across IDL matches and identifying patterns in IDL fan voting.

Methods

Dataset

Event metadata, judges' scores, and fan votes of the 2026 season were scraped from the IDL website (International Dance League, 2026d). The dataset includes 15 Round 1 matches and 5 Round 2 matches.

IDL streams on two YouTube channels, @idl.global (70K followers; International Dance League, n.d.-a) and @steezy (1.97 million followers; STEEZY Studio, n.d.). Live chat replay must be enabled in order to access live chat comments after a stream ends in Youtube. Since IDL did not enable this feature for most of its livestreams, our live chat data is limited.

A total of 23,238 live chat comments were scraped.

Table 1. YouTube Live Chat Comments by Channel

Event	YouTube channel	# of Comments
Sydney	@steezy	411
Seoul	@idl.global	19,775
Seoul	@steezy	2,517
Los Angeles	@steezy	226
Championship	@steezy	309

The @idl.global livestream of the Seoul event accounts for most of the dataset, with 19,775 of the 23,238 comments. The @steezy channel drew far fewer comments, ranging from 226 to 2,517 per event.

Comments were restricted to those posted during each team's performance window so that each comment could be attributed to a specific routine. Comments posted before, between, or after sets were excluded, leaving 3,989 comments (17.2% of the total).

Table 2. YouTube Live Chat Comments by Team

Team	# of Comments
Sydney	
BROTHERHOOD	17
GRV	13
JAM REPUBLIC	6
1MILLION	5
ROYAL FAMILY	5
QUICK STYLE	1
Seoul	
1MILLION	969
BROTHERHOOD	914
QUICK STYLE	613
ROYAL FAMILY	610
JAM REPUBLIC	470
GRV	285
Los Angeles	
BROTHERHOOD	31
GRV	9
JAM REPUBLIC	8
1MILLION	5
QUICK STYLE	1
ROYAL FAMILY	1
Championship	
BROTHERHOOD	13
GRV	7
JAM REPUBLIC	3
1MILLION	1
QUICK STYLE	1

Comments posted while the team was performing, combined across both channels. Comments before, between and after sets are excluded.

At the Sydney, Los Angeles, and Championship events, each team received between 1 and 31 comments, compared to between 285 and 969 comments at the Seoul event. Teams with few comments would have a net sentiment that reflects only a handful of viewers which would not provide a reliable basis for comparison with fan vote share and judges' scores. The analysis of agreement between net sentiment, fan votes, and judges' scores is therefore restricted to the Seoul event with 3861 comments.

Sentiment Classification

IDL YouTube live chat comments differ substantially from standard English. They include shoutouts to teams and dancers, dance-specific slang, spelling errors, and abbreviations. An existing sentiment classification model, AmaanP314/youtube-xlm-roberta-base-sentiment-multilingual (AmaanP314, 2024), is used as the baseline. The model was pre-trained on approximately 198

million multilingual tweets and fine-tuned on over 1 million YouTube comments.

A sample of 500 live chat comments was manually labelled with sentiment. The sample was drawn using stratified sampling by team to prevent over-representation of teams with higher comment volume.

Comments were labelled as positive, negative, or neutral based on the sentiment expressed during each team's performance. Labelling was performed by two labellers with a dance background, chosen for their familiarity with dance terminology, abbreviations, and slang. They agreed on 81% of comments, with a Cohen's kappa of 0.69. Only two comments received opposite labels (positive and negative).

Limitations of this process include subjectivity in judging mixed or ambiguous comments, particularly short exclamations, and possible loss of slang meaning in translated comments. The labels also reflect the interpretations of only two labellers and may not generalize to broader audience consensus.

Baseline Model Evaluation

The baseline model was evaluated on the labelled sample. It achieved an accuracy of 0.479, a macro F1 score of 0.470, and a macro recall of 0.513. The model had low recall for the positive class (0.412), missing a large share of positive comments. It also had low precision for the neutral class (0.297), frequently predicting neutral when the true label was positive or negative. Performance on negative comments was moderate, with similar precision (0.449) and recall (0.460).

Overall, the baseline model struggled to distinguish neutral sentiment from positive or negative sentiment.

Fine-Tuned Model Evaluation

The baseline model was fine-tuned on the labelled sample to improve performance on IDL live chat comments. The fine-tuned model achieved a mean macro F1 of 0.649, an accuracy of 0.705, and a macro recall of 0.652. All three metrics improved substantially over the baseline, indicating that fine-tuning on IDL-specific comments improved the model's ability to distinguish sentiment classes.

The fine-tuned model was then used to label the sentiment of live chat comments from the Seoul event.

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